

TABLE VI
Frequency Coverage Broadband
Antennae Alongwith Desirable
Antenna Factor and Gain in dB

Ant. type	Freq. range	AF (dB)	Gain (dB)
Biconical	30-300 MHz	>10	1 - 4
Log-periodic	300 MHz to 1 GHz	15 - 20	4 - 8
Dual ridgeguide	1 GHz to 18 GHz	>25	>15

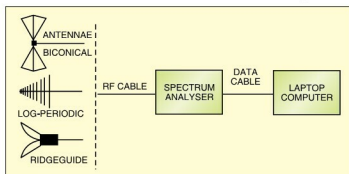


Fig. 16: Typical E-field measurement setup

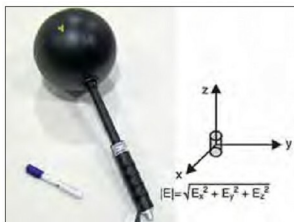


Fig. 17: Isotropic antenna AT3000 (passive probe, 20-3000 MHz)

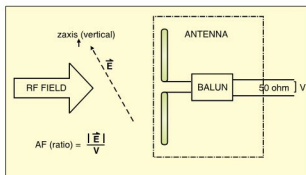


Fig. 18: Antenna factor concept

A typical test setup for electric field measurement is shown in Fig. 16. Although no single technique or test

equipment is suitable to cover the entire frequency band of Table V, a number of reputed manufacturers (HP, Narda, Rohde & Schwarz, etc) market suitable spectrum analysers with optional accessories such as antennae with detector probe to cover a broad band of frequencies of interest. The three types of broadband antennae, as shown in Fig. 16, cover a large enough range of frequencies as shown in

Table VI.

Tri-axial antenna probe. Another type of antenna/probe that is increasingly being favoured for use with spectrum analyser is the tri-axial isotropic antenna/probe, which is made by three independent broadband sensing

elements placed orthogonal to each other. It simplifies the measurement procedure that ideally determine the RMS field strength independently of direction and polarisation of the radiation source since it affords measurement of E-field components along all three axes (X, Y and Z). In fact, the output of each element is measured in three consecutive time intervals, assuming the field components stationary for the measurement duration. This does away with the need for orienting the antenna in the plane of incident electric field.

Detectors. Two main types of detectors generally used with these antennae are thermocouples and Schottky diodes; bolometric devices are rarely used. All three respond to square of voltage induced into the

antenna and hence they can be directly converted to equivalent power density. Schottky diodes can also convert

the RMS value of the detected field components into a DC voltage. By using diodes instead of thermocouples, it is possible to handle a much wider dynamic range of field strength (typically, 60 dB). Thermocouples can also get damaged more easily than diodes at high power levels.

Spectrum analyser. The most commonly used spectrum analysers for this function are the ones wherein the input signal is swept through the selected frequency span, sampling all signal components successively in time.

E-field strength. The RMS voltage from the sensor (antenna + diode detector) in conjunction with spectrum analyser is converted into equivalent electric field strength (RMS) value using equation:

$$E = AF \times V$$

where 'E' is the electric field strength in volts/metre, 'V' is the voltage received by the spectrum analyser and 'AF' is the antenna factor. The unit of AF is 1/metre. The antenna factor is antenna-specific and normally supplied by the manufacturer.

$E = AF \times V$ in dB becomes E (dBV/m) = AF (dB)/m + V (dB).

For example, if the voltage received is -30 dBV and AF is 10 dB/m, then E is -20 dBV/m.

Antenna factor. It is defined as the ratio of incident electric field strength (upon an antenna) to the voltage that is produced at the antenna's terminals into a 50-ohm load (see Fig. 18). The antenna factor is simply a ratio with unit of 1/m; since 'E' is expressed in V/m units, $AF = E/V = V/m \div V = 1/m$.

Antenna factor can also be derived from the knowledge of antenna gain by using the following relationship:

$$AF = E/V = (9.73) / (G^{0.5} \lambda)$$

The above is the ratio version with the dimension of 1/m. (Note that gain (G) is a ratio in the above equation and not dB; for example, $G=1.64$ for a dipole.) The more commonly used logarithmic version with the AF dimensions of dB/m and G as ratio is: AF (in dB/m) = $20 \log [9.73 / (G^{0.5} \lambda)]$ ●